Foundations of ADCS

3. Inorbit Models

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Foundations of ADCS

https://github.com/risherlock/ADCS

0. Course introduction
1. Preliminaries of ADCS
2. Attitude determination system
3. **Inorbit models**
4. Attitude control system
5. ADCS simulation
6. Discussions and conclusion

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    c. Time conversions
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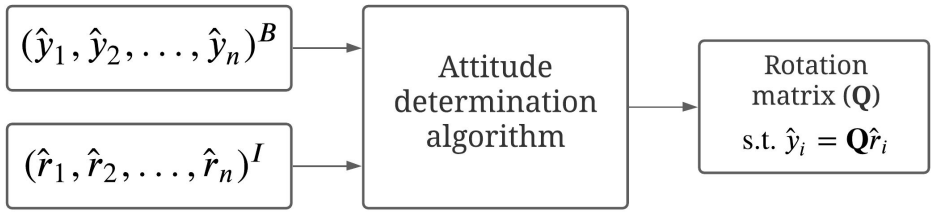
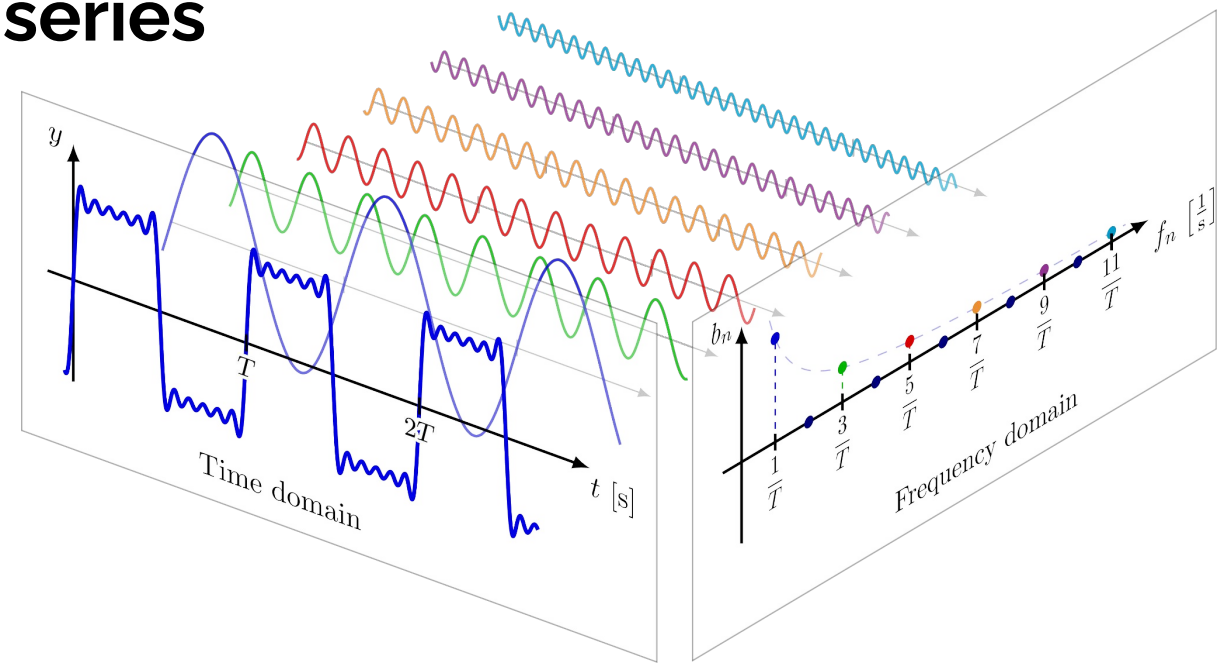
Review: Attitude Determination



Attitude determination is the process which takes a set of vectors in *body frame* and corresponding vectors in *reference frame* to output attitude parameters.

Environment models

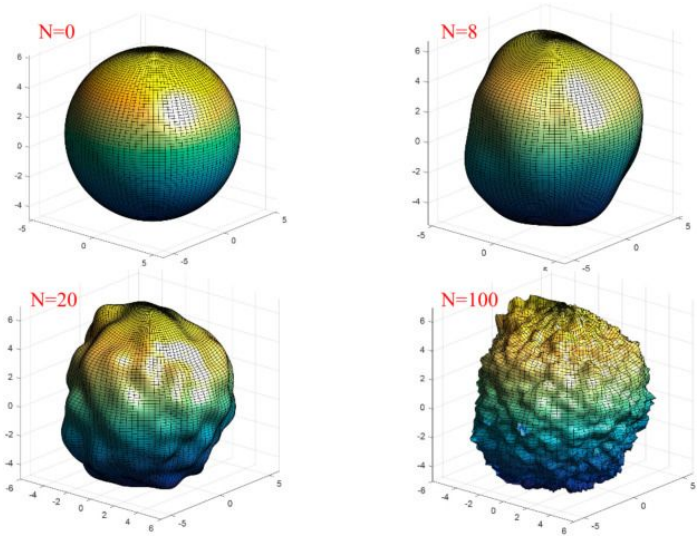
Fourier series

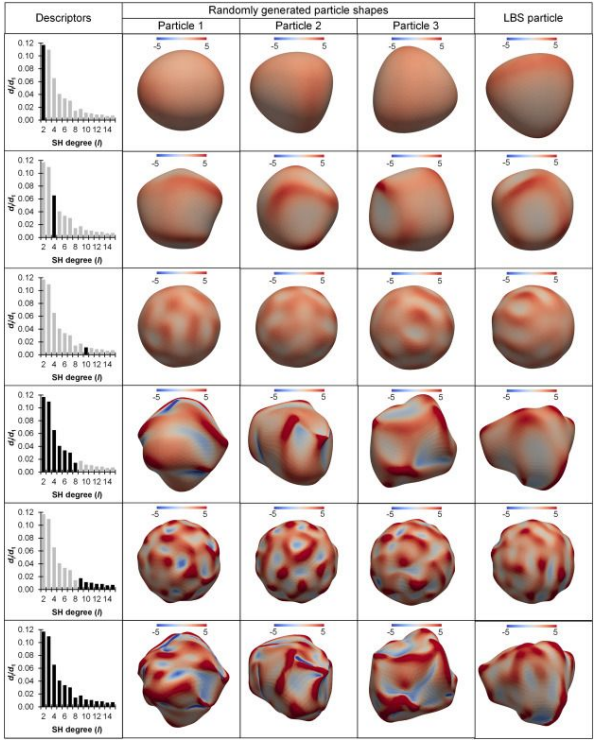


An arbitrary periodic function is sum of trigonometric functions.

Source: https://tikz.net/fourier_series/

Spherical harmonics

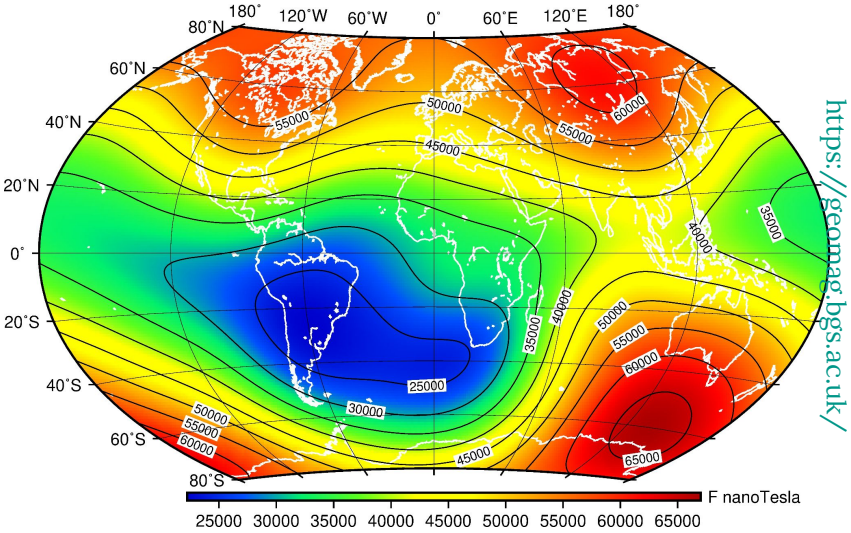




Spherical harmonics is the generalization of the idea of Fourier series on the surface of 2-sphere. A function on spherical manifold can be approximated using spherical harmonic series.

Source: left, right

IGRF model



https://geomag.bgs.ac.uk/

The IGRF is essentially a set of Gaussian coefficients that are put updated every 5 years by IAGA for use in a spherical harmonic model. The model includes both the coefficients for the epoch year and secular variation variables, which track the change of these coefficients in nanoTesla per year. These secular variation coefficients are used to extrapolate the Gaussian coefficients to the date in question.

Source: [1]

IGRF model

Magnetic field, **B**, is defined as the negative gradient of scalar potential function V.

$$\mathbf{B} = -\nabla V$$

A spherical harmonic model of the scalar potential is used to parameterize geomagnetic field as

$$V(r,\theta,\phi) = a\sum_{n=1}^{k}\left(\frac{a}{r}\right)^{n+1}\sum_{m=0}^{n}(g_n^m\cos m\phi + h_n^m\sin m\phi)P_n^m(\theta)$$

where,

- a : reference radius of the Earth (a = 6371.2 km) ✓
- r, θ, φ : radius, co- latitude i.e. (90° - latitude), and longitude respectively ✓
- g, h : Gaussian coefficients released by the IAGA ✓
- P : Schmidt quasi-normalized associated Legendre functions of degree n and order m. n = m = 13 for IGRF-13.

Source: [1]

IGRF model

$$B_r = \frac{-\partial V}{\partial r} = \sum_{n=1}^{k} \left(\frac{a}{r}\right)^{n+2} (n+1) \sum_{m=0}^{n} (g_n^m \cos m\phi + h_n^m \sin m\phi) P_n^m(\theta)$$

$$B_\theta = \frac{-1}{r}\frac{\partial V}{\partial \theta} = -\sum_{n=1}^{k} \left(\frac{a}{r}\right)^{n+2} \sum_{m=0}^{n} (g_n^m \cos m\phi + h_n^m \sin m\phi) \frac{\partial P_n^m(\theta)}{\partial \theta}$$

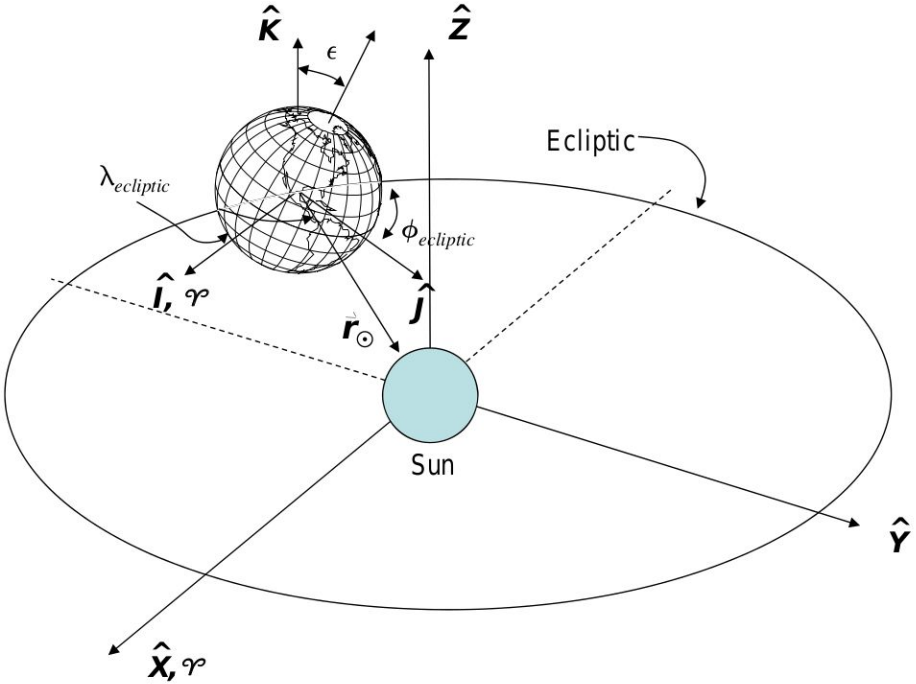
$$B_\phi = \frac{-1}{r\sin\theta}\frac{\partial V}{\partial \phi} = \frac{-1}{\sin\theta} \sum_{n=1}^{k} \left(\frac{a}{r}\right)^{n+2} \sum_{m=0}^{n} m(-g_n^m \sin m\phi + h_n^m \cos m\phi) P_n^m(\theta)$$

B_r, B_θ, and B_φ represent the field strength in local tangential coordinates. Software interface for IGRF would look something like this.

```
[Bn, Be, Bd] = igrf(latitude, longitude, radius, time, NED)
```

Source: [1]

Sun vector model



Geometry for the Sun Position Vector. The solution of the position vector to the Sun rests on determining the ecliptic longitude (ecliptic latitude is 0°) and the range. The *X* and *I* axes are parallel.

Example 5-1: Finding the Sun Position Vector.

GIVEN: April 2, 2006, 00:00 UTC

FIND: Geocentric position vector of the Sun (MOD)

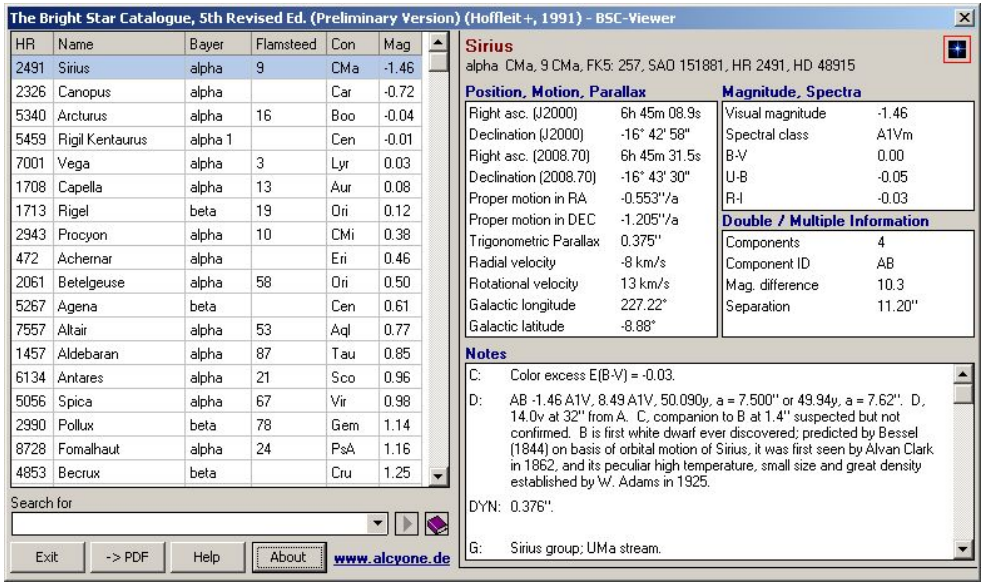
$$\phi_{ecliptic} = 0$$
$$\lambda_{ecliptic} = ?$$
$$\epsilon = ?$$
$$r_\odot = ?$$

$$\mathbf{r}_\odot = r_\odot \cos(\lambda_{ecliptic})\,\mathbf{i} + r_\odot \sin(\lambda_{ecliptic}) \cos(\epsilon)\,\mathbf{j} + r_\odot \sin(\lambda_{ecliptic}) \sin(\epsilon)\,\mathbf{k}$$

Source: [2], [3]

Star tracker

Source: https://doi.org/10.3390/s21030907



Atmospheric torque

$$\mathbf{t}_a = \mathbf{r} \times \mathbf{f}$$
$$\mathbf{f} = -\rho V^2[(2 - \sigma_n - \sigma_t)(\mathbf{e}_v \cdot \mathbf{e}_n)^2\mathbf{e}_n + \sigma_t(\mathbf{e}_v \cdot \mathbf{e}_n)\mathbf{e}_v]dA$$

$\mathbf{t}_a$, $\mathbf{r}$, $\mathbf{f}$: aerodynamic torque, moment arm, and force
ρ : atmospheric density
σ_n, σ_t : normal and tangential momentum exchange coefficient
$\boldsymbol{e}_v$, $\boldsymbol{e}_n$: spacecraft velocity vector and outward normal to dA
dA : projected area of spacecraft element normal to incident flow

Source: [4]

Atmosphere model

Exponentially decaying atmosphere model

$$\rho = \rho_0 \exp\left(-\frac{h - h_0}{H}\right)$$

where,

ρ_0, h_0 : reference density and height

h : height above ellipsoid

H : scale height

Decent results and computationally efficient but not good for high fidelity simulations.

h (km)	h_0 (km)	ρ_0 (kg/m^3)	H (km)
0–25	0	1.225	8.44
25–30	25	3.899×10^{-2}	6.49
30–35	30	1.774×10^{-2}	6.75
35–40	35	8.279×10^{-3}	7.07
40–45	40	3.972×10^{-3}	7.47
45–50	45	1.995×10^{-3}	7.83
50–55	50	1.057×10^{-3}	7.95
55–60	55	5.821×10^{-4}	7.73
60–65	60	3.206×10^{-4}	7.29
65–70	65	1.718×10^{-4}	6.81
70–75	70	8.770×10^{-5}	6.33
75–80	75	4.178×10^{-5}	6.00
80–85	80	1.905×10^{-5}	5.70
85–90	85	8.337×10^{-6}	5.41
90–95	90	3.396×10^{-6}	5.38
95–100	95	1.343×10^{-6}	5.74
100–110	100	5.297×10^{-7}	6.15
110–120	110	9.661×10^{-8}	8.06
120–130	120	2.438×10^{-8}	11.6
130–140	130	8.484×10^{-9}	16.1
140–150	140	3.845×10^{-9}	20.6
150–160	150	2.070×10^{-9}	24.6
160–180	160	1.224×10^{-9}	26.3
180–200	180	5.464×10^{-10}	33.2
200–250	200	2.789×10^{-10}	38.5
250–300	250	7.248×10^{-11}	46.9
300–350	300	2.418×10^{-11}	52.5
350–400	350	9.158×10^{-12}	56.4
400–450	400	3.725×10^{-12}	59.4
450–500	450	1.585×10^{-12}	62.2
500–600	500	6.967×10^{-13}	65.8
600–700	600	1.454×10^{-13}	79.0
700–800	700	3.614×10^{-14}	109.0
800–900	800	1.170×10^{-14}	164.0
900–1,000	900	5.245×10^{-15}	225.0
>1,000	1,000	3.019×10^{-15}	268.0

Source: [5]

Solar radiation pressure torque

Torque produced by the impact of sunlight photons on the surface of the spacecraft.

$$\mathbf{t}_s = \mathbf{r} \times \mathbf{f}$$
$$\mathbf{f} = F_n \mathbf{n} + F_t \mathbf{t}$$

where,

$$F_n = PA\left[(1+\rho_s)\cos^2(\alpha) + \frac{2}{3}\rho_d \cos(\alpha)\right]$$
$$F_t = PA(1-\rho_s)\cos(\alpha)\sin(\alpha)$$

$\mathbf{t}_a$, $\mathbf{f}$: solar radiation torque and force causing it
$\mathbf{r}$: vector from the body center of mass to the optical center of pressure
$\mathbf{n}$: unit vector normal to the surface and opposite to the vector of incoming photons q
$\mathbf{t}$: transverse unit vector perpendicular to the n and in the plane spanned by **q** and **n**
F_n, F_t : components of force in the **n** and **t** directions

Source: [3]

Magnetic torque

Torque produced by the impact of sunlight photons on the surface of the spacecraft.

$$\mathbf{t}_m = \mathbf{m} \times \mathbf{r}_m$$
$$\mathbf{r}_m = -grad(V)$$

where,

$\mathbf{r}_m$: the sum of individual magnetic moment caused by permanent and induced magnetism and the spacecraft-generated current loops

$\mathbf{m}$: geocentric magnetic flux density, Wb/m^2

V : unit vector normal to the surface and opposite to the vector of incoming photons q

Source: [4]

Gravity model

$$\mathbf{f} = \frac{G m_1 m_2 \mathbf{r}}{|\mathbf{r}|^3}$$

$$\mathbf{t}_g = \begin{bmatrix} 6\omega_0^2 (J_{33} - J_{22}) q_1 \\ 6\omega_0^2 (J_{33} - J_{11}) q_2 \\ 0 \end{bmatrix}$$

where,

$\mathbf{J}$: diag(J_{11}, J_{22}, J_{33}) is the moment of inertia

ω_0 : angular orbital velocity

The difference in forces acting on each masses of the dumbbell produces a torque $\mathbf{t}_g$. This torque tends to steer the dumbbell to align it lengthwise with local vertical.

Even of the masses are equal, one can still develop substantial gravity gradient torques on the satellites.

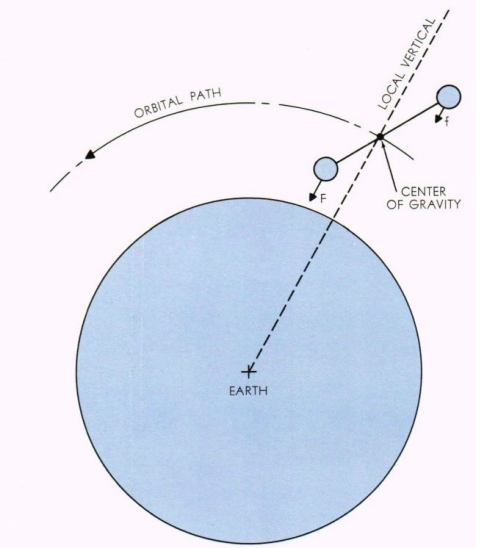
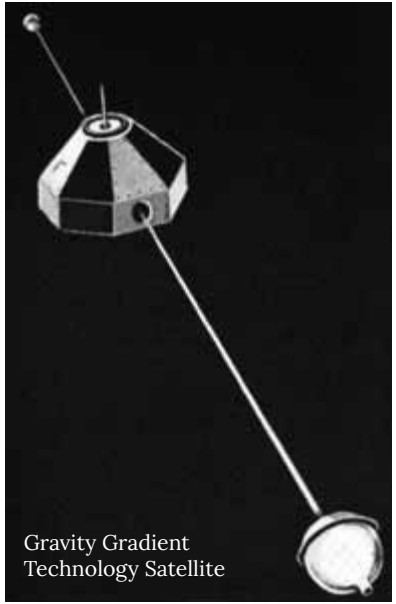


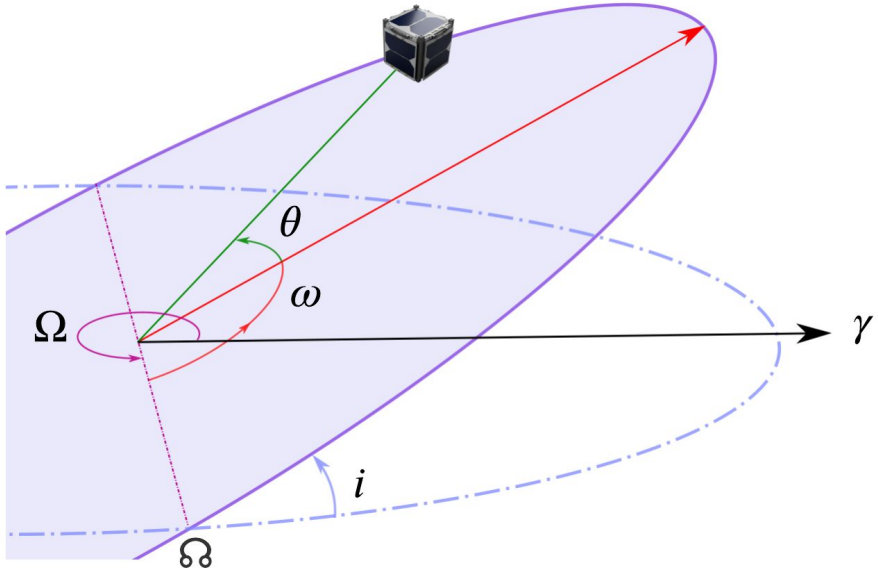
Fig. 1—Gravitational forces acting on a dumbbell-shaped satellite.



Source: [4], [6]

Tools and utilities

Keplerians



Ω	longitude of ascending node
☊	ascending node
i	inclination
♈	vernal equinox
ω	argument of periapsis
θ	true anomaly

TLE

```
ISS (ZARYA)
1 25544U 98067A   08264.51782528 -.00002182  00000-0 -11606-4 0  2927
2 25544  51.6416 247.4627 0006703 130.5360 325.0288 15.72125391563537
```

Field	Columns	Content	Example
1	01	Line number	1
2	03–07	Satellite catalog number	25544
3	08	Classification (U: unclassified, C: classified, S: secret) [11]	U
4	10–11	International Designator (last two digits of launch year)	98
5	12–14	International Designator (launch number of the year)	067
6	15–17	International Designator (piece of the launch)	A
7	19–20	Epoch year (last two digits of year)	08
8	21–32	Epoch (day of the year and fractional portion of the day)	264.51782528
9	34–43	First derivative of mean motion; the ballistic coefficient [12]	-.00002182
10	45–52	Second derivative of mean motion (decimal point assumed) [12]	00000-0
11	54–61	*B**, the drag term, or radiation pressure coefficient (decimal point assumed) [12]	-11606-4
12	63–63	Ephemeris type (always zero; only used in undistributed TLE data) [13]	0
13	65–68	Element set number. Incremented when a new TLE is generated for this object.[12]	292
14	69	Checksum (modulo 10)	7

Source: https://en.wikipedia.org/wiki/Two-line_element_set

TLE

```
ISS (ZARYA)
1 25544U 98067A   08264.51782528 -.00002182  00000-0 -11606-4 0  2927
2 25544  51.6416 247.4627 0006703 130.5360 325.0288 15.72125391563537
```

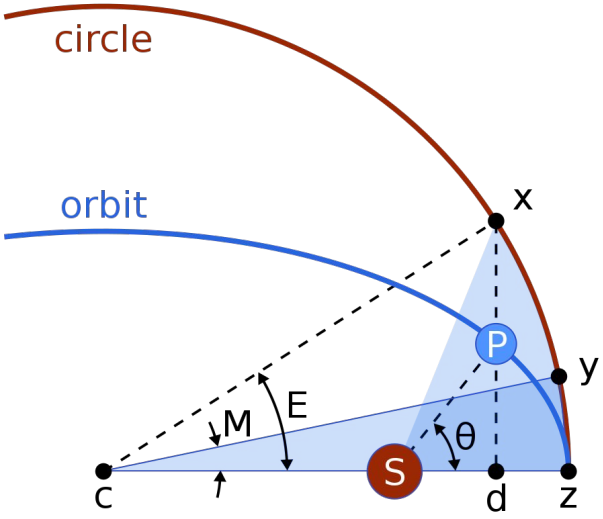
Field	Columns	Content	Example
1	01	Line number	2
2	03–07	Satellite Catalog number	25544
3	09–16	Inclination (degrees)	51.6416
4	18–25	Right ascension of the ascending node (degrees)	247.4627
5	27–33	Eccentricity (decimal point assumed)	0006703
6	35–42	Argument of perigee (degrees)	130.5360
7	44–51	Mean anomaly (degrees)	325.0288
8	53–63	Mean motion (revolutions per day)	15.72125391
9	64–68	Revolution number at epoch (revolutions)	56353
10	69	Checksum (modulo 10)	7

Orbit propagator

$$M = E - e \sin E$$

where,
M : mean anomaly
E : eccentric anomaly
e : Eccentricity

Eccentric anomaly is computed from above equation using numerical methods for it contains E inside transcendental function. True anomaly θ, is computed from E.



Geometric construction for Kepler's calculation of θ. The Sun (located at the focus) is labeled S and the planet P. The auxiliary circle is an aid to calculation. Line xd is perpendicular to the base and through the planet P. The shaded sectors are arranged to have equal areas by positioning of point y.

Time

Solar time: based on the interval between successive transits of the Sun over a local meridian.

Sidereal time: time between successive transits of stars over a particular meridian.

Universal time: time based on a fictitious mean Sun exhibiting uniform motion in right ascension along the equator. Mathematically, UT is derived from sidereal time.

Dynamic time: based on motion about the bodies, such as Moon's motion about the motion about time. It depends on the fact that the time is the independent variable in the quotations of motion describing the object's motion.

Atomic time: precise time standard based on based on the specific quantum transition of electrons in a cesium-133 atom.

Time

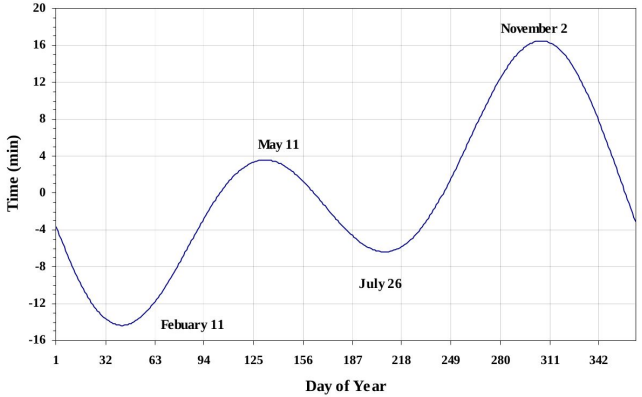


Figure 3-22. Equation of Time Variation. The difference in mean and apparent (or true) solar time varies throughout the year from about plus 16 minutes, to about minus 14 minutes.

The mutual relation of the different time scales and their historical evolution is outlined in the figure in the right

Distinction is made between **dynamical time scales** that serve as independent argument in the equations of motion, **atomic time scales** that provide the practical realization of a uniform clock, and the non-uniform **solar time scales** that are tied to the motion of the Sun and the rotation of the Earth.

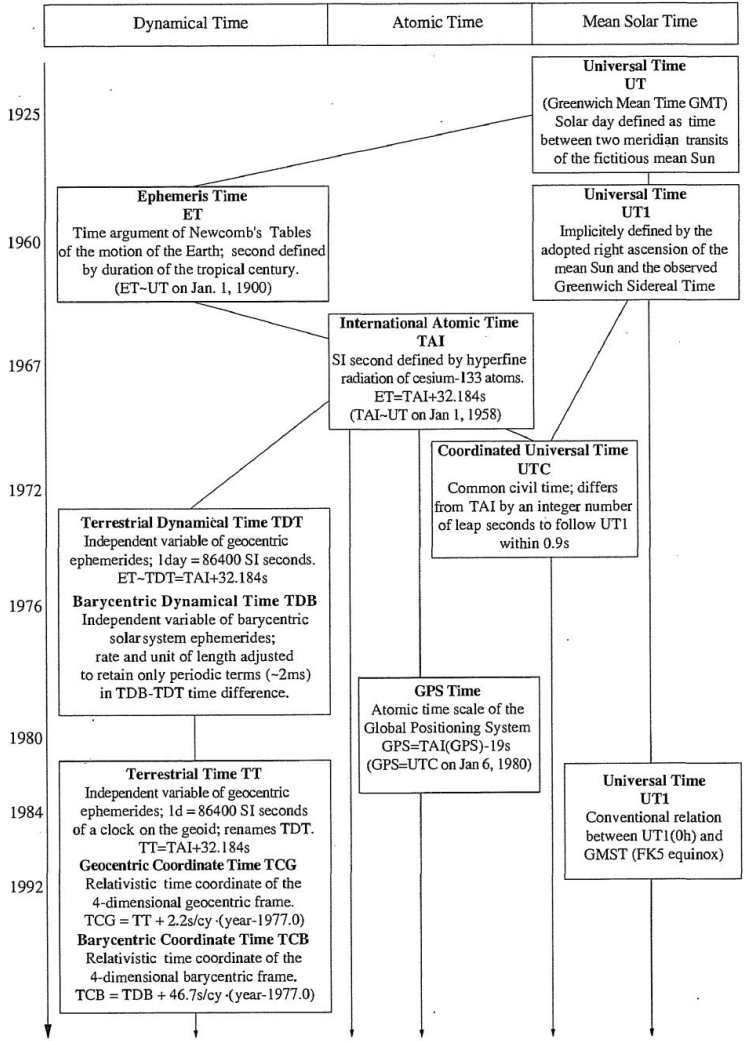


Fig. Evolution of conventional time scales

Source: [1]

Julian date

The Julian date, JD, is the interval of time measured in days from the epoch January 1, 4713 B.C., 12:00.

The JD provides a continuous, simple, concise method of preserving year-month-day-hour-minute-second information in one variable, which is especially nice for digital implementations. In particular, Julian date makes the computation of elapsed days between two events easier in software.

Additionally, many relations for astronomical equations of motion use the number of Julian centuries from a particular epoch eg. J2000.0 which is a Julian Day for 1 Jan, 2000 at 12:00 noon.

Julian date

Because the starting point or reference epoch is so long ago, numbers in the Julian day can be quite large and cumbersome.

A more recent starting point is sometimes used, for instance by dropping the leading digits, in order to fit into limited computer memory with an adequate amount of precision. In the following table, times are given in 24-hour notation.

Name	Epoch	Calculation	Current value	Notes
Julian date	12:00 January 1, 4713 BC proleptic Julian calendar	JD	**2460078.63333**	
Reduced JD	12:00 November 16, 1858	JD − 2400000	**60078.63333**	[13][14]
Modified JD	0:00 November 17, 1858	JD − 2400000.5	**60078.13333**	Introduced by SAO in 1957
Truncated JD	0:00 May 24, 1968	floor (JD − 2440000.5)	**20078**	Introduced by NASA in 1979
Dublin JD	12:00 December 31, 1899	JD − 2415020	**45058.63333**	Introduced by the IAU in 1955
CNES JD	0:00 January 1, 1950	JD − 2433282.5	**26796.13333**	Introduced by the CNES[15]
CCSDS JD	0:00 January 1, 1958	JD − 2436204.5	**23874.13333**	Introduced by the CCSDS[15]
Lilian date	day 1 = October 15, 1582[b]	floor (JD − 2299159.5)	**160919**	Count of days of the Gregorian calendar
Rata Die	day 1 = January 1, 1[b] proleptic Gregorian calendar	floor (JD − 1721424.5)	**738654**	Count of days of the Common Era
Mars Sol Date	12:00 December 29, 1873	(JD − 2405522)/1.02749	**53096.93229**	Count of Martian days
Unix time	0:00 January 1, 1970	(JD − 2440587.5) × 86400	**1684033924**	Count of seconds,[16] excluding leap seconds
.NET DateTime	0:00 January 1, 1 proleptic Gregorian calendar	(JD − 1721425.5) × 864000000000	**6.3819630723997E+17**	Count of 100-nanosecond ticks, excluding ticks attributable to leap seconds[17]

Fig. Julian date variants

Source: https://en.wikipedia.org/wiki/Julian_day

Coordinate transformations

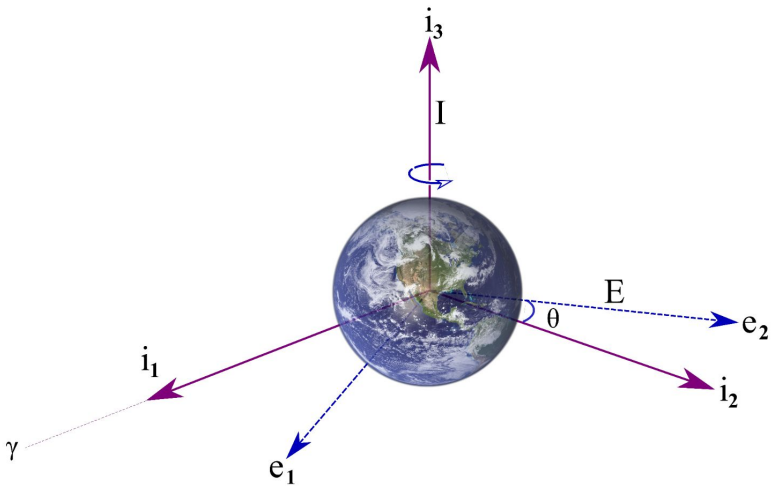


Fig. Earth Centered Inertial frame (I) and Earth Centered Earth Fixed frame (E)

The ECEF is rotated from the ECI frame in z-axis only, and the DCM from ECI to ECEF is

$$Q_E^I = \begin{bmatrix} cos\theta & sin\theta & 0 \\ -sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

References

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3. Howard Curtis - *Orbital Mechanics For Engineering Students* (2020)
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5. Landis Markley, John Crassidis - *Fundamentals of Spacecraft Attitude Determination and Control* (2014)
6. R. E. Fischell - *Gravity Gradient Stabilization of Earth Satellites*
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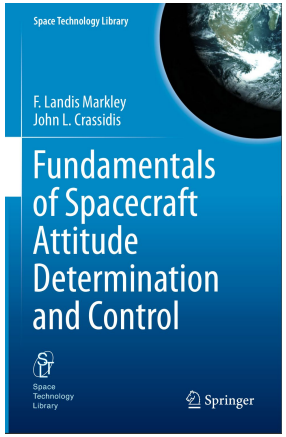
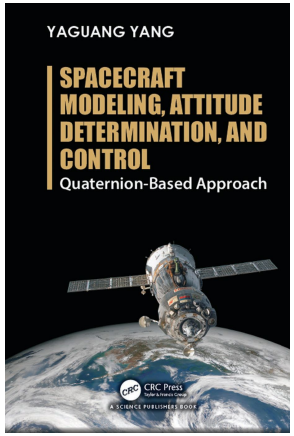
Books


Space Technology Library
F. Landis Markley
John L. Crassidis
Fundamentals of Spacecraft Attitude Determination and Control
Space Technology Library
Springer


YAGUANG YANG
SPACECRAFT MODELING, ATTITUDE DETERMINATION, AND CONTROL
Quaternion-Based Approach
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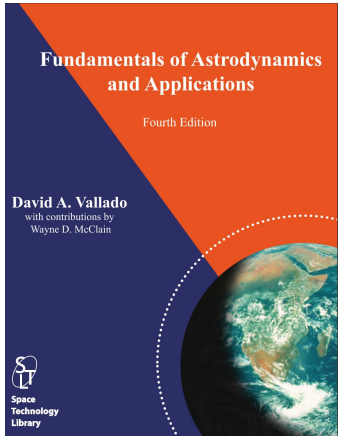
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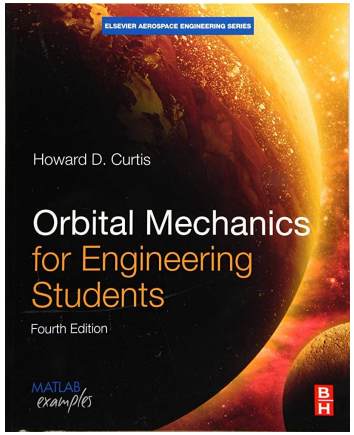
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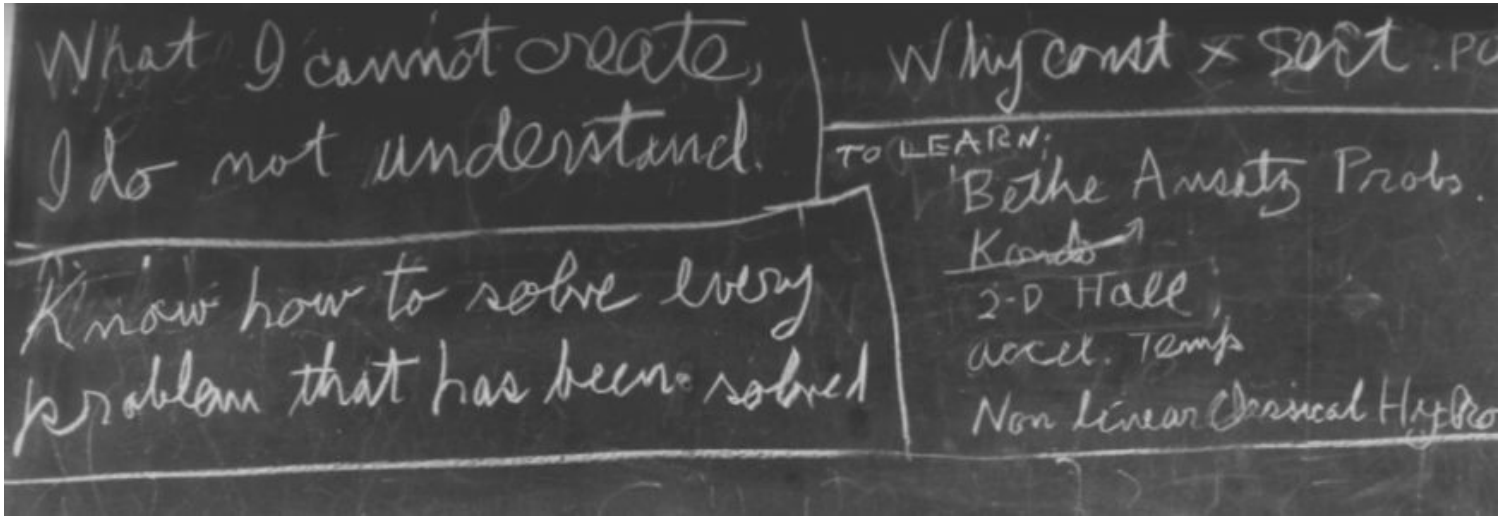
Orbital Mechanics for Engineering Students
Fourth Edition
MATLAB examples



Richard Feynman's blackboard at the time of his death, 1988.

Thank you!